

Floating point exception in torch.mkldnn_max_pool2d #149274

Closed



default1360 opened on Mar 16, 2025 · edited by pytorch-bot

Edits ...

Describe the bug

Running the following PyTorch code results in a Floating point exception crash:

```
import torch

x = torch.randn(2, 64, 32, 32).to_mkldnn()
out2 = torch.mkldnn_max_pool2d(x, kernel_size=3, stride=0)
```

Versions

PyTorch version: 2.6.0+cu124
Is debug build: False
CUDA used to build PyTorch: 12.4
ROCM used to build PyTorch: N/A

cc @malfet @gujinghui @PenghuiCheng @XiaobingSuper @jianyuh @jgong5 @mingfeima @sanchitintel @ashokei @jingxu10 @min-jean-cho @yanbing-j @Guobing-Chen @Xia-Weiwen @snadampal

CaoE self-assigned this on Mar 17, 2025

zou3519 added triaged module: mkldnn on Mar 17, 2025



carnil on Apr 2, 2025 ...

[CVE-2025-2953](#) is assigned for this issue.



CaoE on Apr 2, 2025 Collaborator ...

Hi [@default1360](#) stride=0 is an invalid setting for maxpool. Is there any practical cases for such usage ?



malfet added [module: error che...](#) [module: empty te...](#) [topic: fuzzer](#) on Apr 22, 2025



malfet assigned [malfet](#) and unassigned [CaoE](#) on Apr 22, 2025

malfet on Apr 22, 2025

Contributor ...

Fixeable by something like the following:

```
diff --git a/aten/src/ATen/native/mkldnn/Uutils.cpp b/aten/src/ATen/native/mkldnn/Uutils.cpp
index 8c74d23e936..561193a959a 100644
--- a/aten/src/ATen/native/mkldnn/Uutils.cpp
+++ b/aten/src/ATen/native/mkldnn/Uutils.cpp
@@ -19,6 +19,7 @@ std::vector<int64_t> pool_output_sizes(
    output_size[1] = input_size[1];

    for (const auto i : c10::irange(2, input_size.size())) {
+    TORCH_CHECK_VALUE(stride[i - 2] > 0, "Strides must be positive!");
    output_size[i] = pooling_output_shape_pad_lr<int64_t>(
        input_size[i],
        kernel_size[i - 2],
```



malfet mentioned this on Apr 22, 2025

[\[MKLDNN\] Check that strides are positive #151848](#)



malfet added a commit that references this issue on Apr 22, 2025

[\[MKLDNN\] Check that stirdes are postive](#) ...

64b3ee5



pytorchmergebot closed this as [completed](#) in [6f32712](#) on Apr 22, 2025



pull added a commit that references this issue on Apr 22, 2025

[\[MKLDNN\] Check that strides are positive \(pytorch#151848\)](#) ...

6f32712



atalman added this to the [2.7.1](#) milestone on Apr 22, 2025



pytorchbot added a commit that references this issue on May 8, 2025

[\[MKLDNN\] Check that strides are positive \(#151848\)](#) ...

8d69296



pytorchbot mentioned this on May 8, 2025

[\[MKLDNN\] Check that strides are positive #153092](#)



malfet added a commit that references this issue on May 8, 2025

[\[MKLDNN\] Check that strides are positive \(#153092\)](#) ...

[Verified](#) 27e9ca5



yeshsurya mentioned this on Aug 5, 2025

 [\[vfix\]: update torch in base image and docker runs for other environm... Azure/azureml-assets#4371](#)



depthfirst-dev mentioned this in 4 pull requests [on Oct 24, 2025](#)



[\[depthfirst-5905\]](#) Upgrade multiple packages in uv.lock seshanktest/tts#1



[\[depthfirst-5909\]](#) Upgrade multiple packages in uv.lock seshanktest/tts#2



[\[depthfirst-5911\]](#) Upgrade multiple packages in uv.lock seshanktest/tts#3



[\[depthfirst-5913\]](#) Upgrade multiple packages in uv.lock seshanktest/tts#4

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Assignees



malfet

Labels

module: empty tensor

module: error checking

module: mkldnn

topic: fuzzer

triaged

Type

No type

Projects

No projects

Milestone



2.7.1

Closed on Aug 14, 2025, 97% complete

Relationships

None yet

Development



Code with agent mode



No branches or pull requests

Participants



+1

